

MAXIM PIATINE

max.piatine@hotmail.com

438-830-8284

github.com/maxpiatine

<https://maxpiatine.com/>

Applied Machine Learning Engineer who builds models and ships them. Combines production backend and cloud-infrastructure experience — deploying and operating services for **100,000+** users — with hands-on **PyTorch** modeling and a computational-science background spanning physics, statistics, and computational neuroscience.

TECHNICAL SKILLS

Languages Python, Golang, C++, SQL, Bash, R, Rust, JavaScript

ML & Scientific Computing PyTorch, spiking neural networks, computer vision, numerical simulation, statistical modeling

Infrastructure & Cloud AWS, Docker, Linux (Ubuntu, RedHat, CentOS), Nginx, Route53, CI/CD

Databases PostgreSQL, MySQL, MSSQL, Oracle, MongoDB; query optimization

Tools & Observability Git, Grafana, REST & RPC APIs, gorilla/mux

Spoken English & French (bilingual), Russian

EXPERIENCE

Software Engineer, NRT Technology

Toronto, ON (May 2024 - Present)

- Led database integration and query optimization for a data-driven drawing system across multiple properties, reverse-engineering a **700-table schema** to identify critical data sources. Wrote and tuned SQL queries handling thousands of daily events, achieving **sub-5-second** return times for multiple months of data.
- Deployed and operated **30** production backend systems in high-security Linux environments, delivering **Golang** services consumed by mobile applications serving **100,000+** users.
- Migrated latency-sensitive legacy Python services to **Golang** on **AWS**, improving runtime efficiency by **23%** and reducing request latency across production environments.
- Implemented production security and observability features: MFA and JWT-based authentication, OTP identity verification, and centralized logging and metrics in **Grafana** for real-time monitoring.

Research Assistant, University of Ottawa

Ottawa, ON (Sept 2024 - Sept 2025)

- Built and trained **spiking neural networks** in **PyTorch**, analyzing transient amplification effects under direct and indirect cortical stimulation.
- Conducted computational modeling and simulation of cortical microstimulation, investigating excitatory and inhibitory recruitment dynamics through reduced-order biophysical models.
- Performed quantitative analysis by fitting model outputs to experimental data, evaluating consistency between simulated and observed neural responses.

Software Developer / Full Stack Developer, Rolls-Royce

Montreal, QC (June 2022 - June 2023)

- Architected and built a **Django/MVC** Data Management System to centralize engine test data across multiple facilities, ingesting test data directly from engines into MySQL for downstream analytics.
- Implemented an automated file processing pipeline (detection / processing / logging) with real-time directory monitoring, capturing new test runs and logging all errors without data loss.

PROJECTS

Statistical Methods of Machine Learning — Kaggle Competition

- Built an R pipeline for a high-dimensional regression task (112 predictors, 3,000 training observations), training regression-tree models with cross-validation-based pruning for model selection to minimize test RMSE.
- Placed **13th of 136 teams** (top 10%).
- **Technology:** R

Receipt Scanning Application

- Built a **computer vision** application to automatically extract itemized expenses from receipt images, validating receipt authenticity and extracting merchant name, line items, prices, and totals into structured data.
- **Technology:** Python, PyTorch

Expenses & Savings Platform

- Designed and built a full-stack platform that scrapes product and pricing data across multiple grocery chains and uses PostgreSQL full-text search to surface cost-saving opportunities across stores.
- **Technology:** C++, Golang, Supabase (PostgreSQL)

EDUCATION

University of Ottawa, M.Sc. Physics (Thesis)

Ottawa, ON (Sept 2023 - Sept 2025)

- Thesis: *Excitatory and Inhibitory Cable Properties Controls the Transient Amplification of Electrical Stimulation* — biophysical, mean-field, and spiking-network modeling fit to experimental neural responses. [\[thesis\]](#)

University of Toronto, Honours B.Sc.

Toronto, ON (Sept 2018 - April 2022)

- Double major in **Physics & Statistics**, **Mathematics** minor.